

# Cloud IQ – AI-Powered Cloud Cost Optimizer

---

## Test Case Documentation

Team Name: Cloud IQ  
Hackathon: IM-03 – Cloud Cost Optimizer (Mock)

### 1. Project Overview

Cloud platforms make it easy for organizations to scale their infrastructure, but they also introduce a common challenge—managing costs effectively. Over time, many cloud resources become underutilized or completely unused, yet continue to generate monthly expenses.

In most cases, engineering teams only become aware of these inefficiencies after reviewing billing reports. By then, unnecessary spending may have continued for weeks or even months.

Our team wanted to address this problem by building a system that continuously analyzes cloud resource usage, identifies waste, and provides actionable recommendations before costs escalate.

Cloud IQ is an AI-powered cloud cost optimization assistant that helps organizations identify opportunities to reduce cloud spending.

### 2. Scope of Testing

Features Tested:

- CSV billing data ingestion
- Resource utilization analysis
- Idle resource detection
- Oversized resource detection
- AI recommendation generation
- Severity classification
- GitHub issue automation
- Discord notification delivery
- Email report generation
- Dashboard creation
- Weekly trend analysis

### 3. Testing Environment

| Component            | Environment      |
|----------------------|------------------|
| Programming Language | Python 3.11      |
| AI Model             | Ollama (Llama 3) |
| GitHub Integration   | REST API         |

|                         |                    |
|-------------------------|--------------------|
| Notifications           | Discord Webhooks   |
| Email Service           | Gmail SMTP         |
| Dashboard               | HTML & CSS         |
| Browser                 | Google Chrome      |
| Development Environment | Visual Studio Code |

#### 4. Success Criteria

| Metric                          | Target          |
|---------------------------------|-----------------|
| Resource Detection Accuracy     | Above 90%       |
| False Positive Rate             | Below 10%       |
| Dashboard Generation Time       | Under 5 seconds |
| AI Recommendation Response Time | Under 3 seconds |
| GitHub Issue Creation           | Successful      |
| Discord Notification Delivery   | Successful      |
| Email Report Delivery           | Successful      |

#### 5. Test Data

Sample Resource Dataset

| Resource ID | Type | CPU Usage | Monthly Cost |
|-------------|------|-----------|--------------|
| srv-01      | EC2  | 5%        | \$220        |
| srv-02      | EC2  | 85%       | \$180        |
| srv-03      | RDS  | 2%        | \$320        |
| srv-04      | S3   | 0%        | \$95         |
| srv-05      | EC2  | 15%       | \$190        |

Weekly Savings Trend

| Week   | Potential Savings |
|--------|-------------------|
| Week 1 | \$120             |
| Week 2 | \$240             |
| Week 3 | \$405             |
| Week 4 | \$525             |

#### 6. Test Results

| Test ID | Feature Tested               | Outcome |
|---------|------------------------------|---------|
| TC-001  | CSV Data Loading             | Passed  |
| TC-002  | Idle Resource Detection      | Passed  |
| TC-003  | Oversized Resource Detection | Passed  |
| TC-004  | Savings Estimation           | Passed  |
| TC-005  | AI Recommendation Generation | Passed  |
| TC-006  | Severity Classification      | Passed  |
| TC-007  | AI Fallback Mechanism        | Passed  |
| TC-008  | GitHub Issue Creation        | Passed  |
| TC-009  | Duplicate Issue Prevention   | Passed  |
| TC-010  | Discord Alert Delivery       | Passed  |
| TC-011  | Notification Summary         | Passed  |

|        |                                  |        |
|--------|----------------------------------|--------|
| TC-012 | Email Report Generation          | Passed |
| TC-013 | Dashboard Generation             | Passed |
| TC-014 | Trend Analysis<br>Visualization  | Passed |
| TC-015 | End-to-End Workflow<br>Execution | Passed |

All planned test cases were completed successfully.

## 7. End-to-End Workflow Validation

Resources analyzed: 8

Optimization opportunities identified: 3

Estimated monthly savings: \$525

Annual savings potential: \$6,300

GitHub issues created successfully

Notifications delivered successfully

Dashboard generated successfully

## 8. What Makes Cloud IQ Different

AI-powered reasoning using Ollama, historical trend analysis, automated GitHub issue creation, multi-channel notifications, offline AI execution, and fault-tolerant execution.

## 9. Business Impact

Cloud visibility improves significantly. Investigation effort is reduced. Response times are shortened. Cost optimization becomes proactive instead of reactive.